

Automating Risk: Utilising FIFA-Approved Skeletal Tracking Data to Identify Biomechanical Markers of Fatigue and Non-Contact Injury Risk in Elite Football

Keith Joseph Fernandes

Research Proposal · in collaboration with the FIFA Innovation Department

ABSTRACT

This research project investigates the validation and application of FIFA-approved Semi-Automated Offside Technology (SAOT) skeletal tracking data for injury prevention in elite football. By comparing markerless pose estimation against gold-standard Vicon motion capture, the study aims to quantify the “Validation Gap” and identify “Kinematic Decay” signatures associated with fatigue. The final output will be an Explainable AI (XAI) dashboard designed to provide actionable injury-risk insights for elite coaching and medical staff.

1. Introduction

Athlete monitoring in football has traditionally relied on “external load” metrics via GPS^[21]. However, these metrics fail to capture the “internal load”—the physiological and biomechanical stress experienced by the athlete. The emergence of FIFA’s Semi-Automated Off-Side Technology (SAOT) infrastructure, which captures 29 skeletal points at 50 Hz for all players, provides a high-fidelity spatiotemporal record of match play^{[8][9]}.

The primary challenge lies in the “Validation Gap.” Marker-based Vicon systems remain the “Gold Standard” for precision, while markerless video-based pose estimation is susceptible to occlusion and spatial uncertainty^{[5][14]}. This research argues that forensic validation is required before SAOT data can be used for clinical injury prediction.

2. Literature Review

The identification of non-contact injury risk in football has traditionally relied on the Acute:Chronic Workload Ratio (ACWR), which compares a player's recent training volume to their long-term average^[10]. However, recent systematic reviews have highlighted the limitations of these locomotor-based models, noting that they fail to capture the biomechanical “efficiency” of movement^[2]. A player may maintain their high-speed running distance while simultaneously exhibiting signs of “kinematic decay”—subtle changes in joint alignment and movement patterns that precede structural failure^[23].

2.1 The Biomechanics of Non-Contact Injury and Fatigue

Non-contact injuries, particularly to the Anterior Cruciate Ligament (ACL) and hamstring complex, are often the result of “mechanical load” exceeding the structural capacity of the musculoskeletal system^[15]. Fatigue, both physical and mental, is a primary driver of this imbalance. Studies have demonstrated that mental fatigue leads to a more upright landing posture, characterised by decreased knee and trunk flexion, which in turn increases the horizontal backward ground reaction force (GRF) and the valgus moment at the knee joint^[23]. Dynamic Knee Valgus (DKV)—the medial collapse of the knee during weight-bearing activities—is a well-established biomechanical marker for ACL strain^[1].

2.2 Predictive Biomechanical Markers Identified in Literature

- **Trunk Lean:** Lateral trunk displacement increases the abduction moment arm at the knee, exacerbating the risk of ligamentous injury^[23].
- **Hip Adduction:** Fatigue reduces the capacity of the hip abductors to control femoral rotation, leading to excessive medial collapse during cutting manoeuvres^[1].
- **Knee Flexion Velocity:** A reduction in the rate of knee flexion upon landing suggests a compromised energy dissipation strategy, shifting the load from the muscles to the passive joint structures^[23].

2.3 The Validation Gap

The fundamental problem lies in the “Validation Gap.” While markerless camera-based systems have shown promise in industrial ergonomic assessments (with errors ranging from 5–20 mm), their performance in the chaotic environment of a football match remains unquantified^[18].

The positional uncertainty in SAOT systems can be mathematically approximated using standard error propagation. For instance, at a standard 50 Hz capture rate, a sprinting player moving at 8 m/s travels 16 cm between frames^[20]. Combining this with spatial resolution uncertainty ($\Delta x_{\text{spatial}}$) and calibration error ($\Delta x_{\text{calibration}}$), the total positional uncertainty (Δx_{total}) can reach approximately 34 cm^[14]:

$$\Delta x_{\text{total}} \approx \sqrt{\Delta x_{\text{temporal}}^2 + \Delta x_{\text{spatial}}^2 + \Delta x_{\text{calibration}}^2}$$

$$\Delta x_{\text{total}} \approx \sqrt{16^2 + 28^2 + 10^2} \approx 34 \text{ cm}$$

This magnitude of error is potentially significant when trying to calculate joint angles, where a few degrees of deviation in knee valgus can distinguish between a safe landing and an ACL-threatening event^[11]. Therefore, the research must first establish the accuracy limits of FIFA-approved skeletal tracking systems before they can be integrated into a reliable injury-prevention framework.

2.4 Theoretical Framework: Kinematic Decay

The concept of “Kinematic Decay” refers to the progressive loss of movement consistency as a player reaches a state of functional fatigue^[3]. Unlike heart rate or lactate levels, which measure physiological stress, kinematic decay measures the degradation of the motor control system^[22]. As muscles fatigue, their ability to generate force decreases, leading to altered movement patterns and impaired proprioception^[17]. In football, this often manifests as reduced coordination during “kinematic sequencing”—the transfer of force from the legs to the trunk and extremities during actions like passing or shooting^[16]. By identifying the

point at which a player's kinematics deviate from their established baseline, coaches can make data-backed decisions regarding substitutions or training intensity adjustments.

3. Aims and Objectives

The aim of this research is to develop a validated application of skeletal tracking data for identifying fatigue and injury risk in elite football.

3.1 Objectives

1. Compare kinematic descriptors (e.g., knee valgus, hip flexion) derived from SAOT systems against Vicon motion capture (MoCap) data to quantify Root Mean Square Error (RMSE).
2. Identify the “Kinematic Decay” signature, defined as statistically significant changes in skeletal markers as players approach fatigue.
3. Develop an Explainable AI (XAI) “Risk Dashboard” prototype to provide actionable insights for coaches and medical staff.

The primary technical challenge associated with Objective 2 lies in computational efficiency and high-throughput data engineering. Processing 29 skeletal data points for 22 players at a frequency of 50 Hz generates a large, high-dimensional dataset that must be parsed and modelled with minimal latency.

4. Methodology

This research adopts a three-phase approach:

4.1 Phase 1: Laboratory Validation

Research will be anchored in the state-of-the-art facilities of an accredited university sports science institute, featuring a world-leading 3D movement capture suite and a 3 Tesla MRI scanner for detailed musculoskeletal assessment.

Elite players will perform football-specific drills (e.g., cutting, sprinting, landing), tracked simultaneously using a Vicon Vero motion capture system and a SAOT-mimicking camera array^[7].

4.2 Phase 2: Big Data and Anomaly Detection

Using FIFA™ datasets, spatiotemporal Graph Neural Networks (GNNs) will be employed to encode player movements and capture spatial interactions between joints^[6]. DeepHit neural networks will be used to forecast “time-to-injury,” accounting for competing risks and individual baseline variability^[4].

4.3 Phase 3: Stakeholder Integration

Qualitative interviews with professional stakeholders (n = 14–20) will be conducted to evaluate the “Risk Dashboard” using the Technology Acceptance Model (TAM)^[19]. This phase ensures that visualisations are intuitive and effectively integrated into existing Risk Management Protocols (RMPs)^[12].

5. Potential Issues and Limitations

The primary limitation of this research is the environmental variability inherent in stadium-based optical tracking compared to controlled laboratory settings. Factors such as player occlusion, varying lighting conditions, and jersey fit may introduce noise into the skeletal coordinates. Additionally, while Graph Neural Networks (GNNs) offer high predictive power, their computational demands require significant high-throughput data engineering to ensure minimal latency for real-time applications.

6. Ethical and Governance Considerations

This study will involve the processing of highly sensitive biomechanical data from elite professional athletes. Ethical approval will be sought through the host institution’s University Research Ethics Committee (UREC). Data governance will adhere to GDPR and

FIFA Data Protection Regulations, ensuring all skeletal data is de-identified and stored on encrypted, secure servers at the host institution. Informed consent will be obtained from all participants in Phase 1 and Phase 3.

7. Proposed Timeline

- **Months 1–6:** Literature synthesis and UREC ethical approval.
- **Months 7–18:** Phase 1 Laboratory Validation and data collection.
- **Months 19–30:** Phase 2 Big Data modelling and GNN development using FIFA datasets.
- **Months 31–36:** Phase 3 Stakeholder interviews, dashboard prototyping, and thesis submission.

8. Dissemination of Findings

Findings will be shared through peer-reviewed journals (e.g., British Journal of Sports Medicine) and presentations at international conferences such as the FIFA Quality Programme forums. The project will also produce a technical “Risk Dashboard” prototype for direct integration into professional club Risk Management Protocols (RMPs)^[12].

9. Conclusion

This research shifts the FIFA Quality Programme from a focus on hardware durability to one of data-driven clinical utility^[9].

Key Contributions

- **Democratisation:** Smaller member associations can access elite biomechanics through stadium camera systems, removing the need for expensive wearable infrastructure^[8].

- **Athlete Wellbeing:** Early detection of biomechanical “red flags” enables targeted prehabilitation strategies, reducing the risk of catastrophic injuries.
- **Mathematical Risk Modelling:** The study will use XGBSE modelling to estimate the probability of injury $P(T \leq t \mid x)$, enabling dynamic and individualised risk profiling^[13].

Final Remarks

This research programme, developed in partnership with FIFA, seeks to modernise injury management by leveraging SAOT infrastructure. By moving beyond “black box” locomotor data toward a more forensic understanding of “kinematic load,” this research aims to establish a new global standard for protecting the game’s most valuable assets: the players.

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